

Adversarial Validity Checks

Strategy-1 ρ · Text-Attributed Graph Framework

Run: run_clean_20260730 · Scoring: raw_gnn ($S_{\text{GNN_step1}} \times 100$)

Check 1A – Coefficient of Variation (sample stability artifact test)

Check 1B – ρ vs N-samples curve, ArXiv (signal exists at $n = 5$?)

Check 2A – Task_Idx ablation (task-identity confound test)

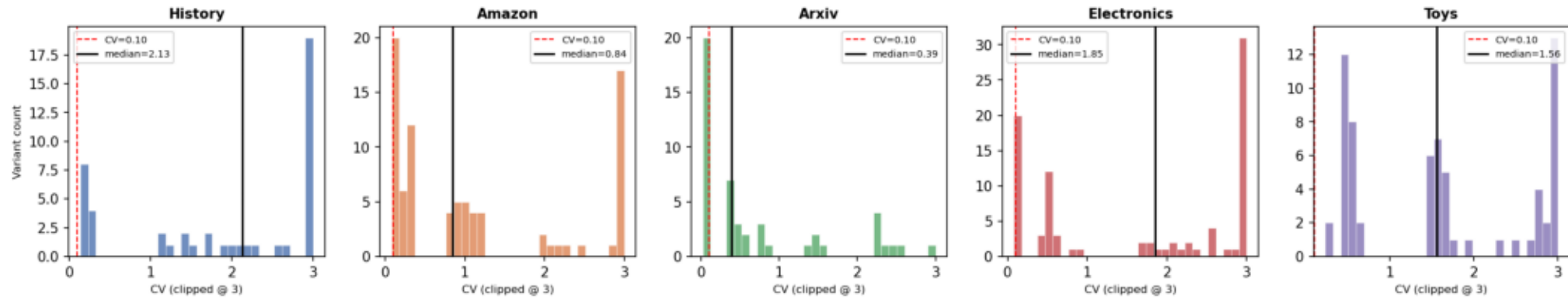
Check 2B – Within-task-type ρ (construction quality signal)

Check 3 – train_mean vs test_mean scatter (confound visualization)

Check 5 – Score compression by task type (Electronics & Toys)

Check 1A — Coefficient of Variation per Dataset

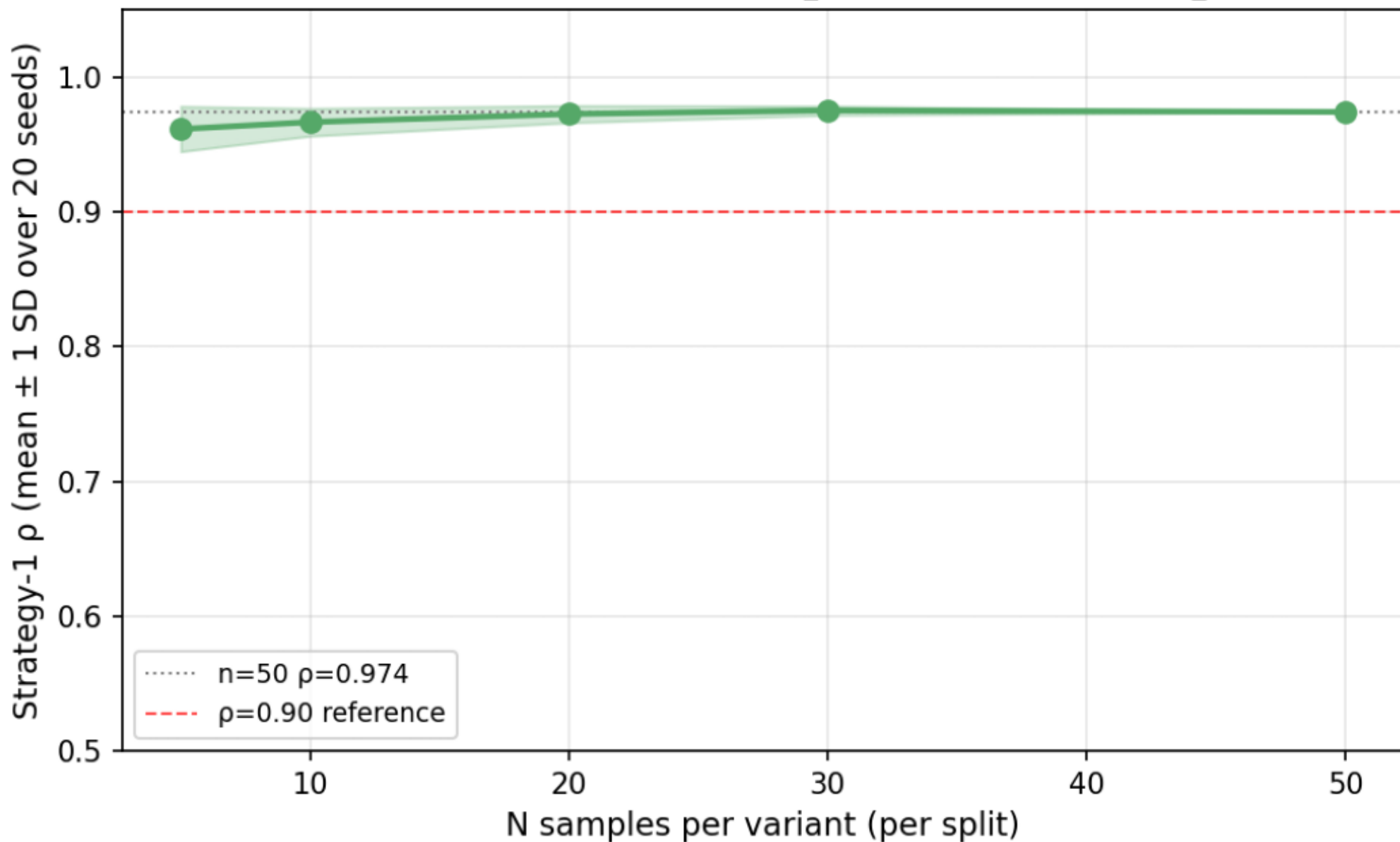
Check 1A — CV Distribution per Dataset (raw_gnn scores, all samples)



CV = $\text{std} / |\text{mean}|$ across all 100 samples per variant. High CV = scores vary substantially across samples; high ρ is not a large-sample LLN artifact. Red dashes = CV 0.10; black line = median CV.

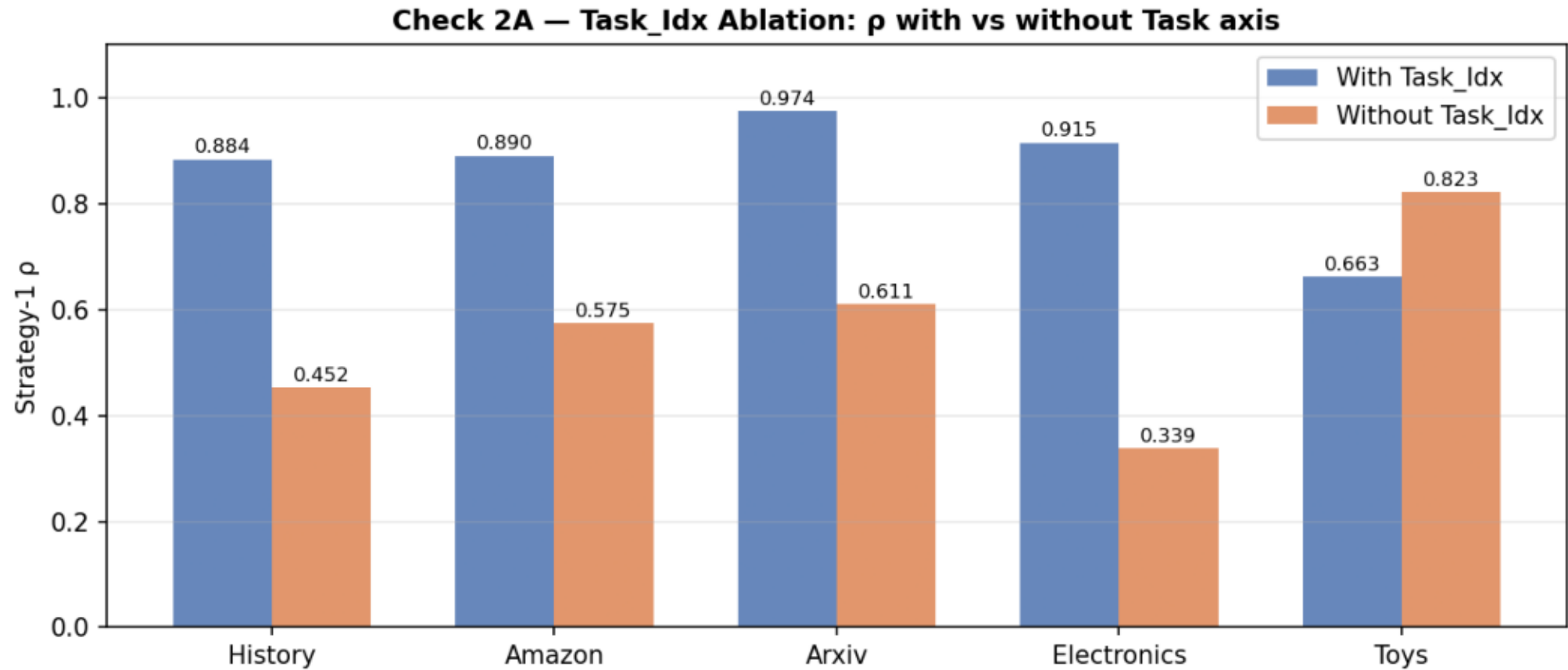
Check 1B — Strategy-1 ρ vs N Samples (ArXiv, raw_gnn)

Check 1B — ArXiv: ρ vs N samples (raw_gnn, zero-filtered, max_depth=4)



Mean $\rho \pm 1$ SD over 20 random subsamples of n train + n test samples per variant (max 50/split).
 $\rho = 0.962$ already at $n = 5$ — the signal is real, not manufactured by having 100 samples.

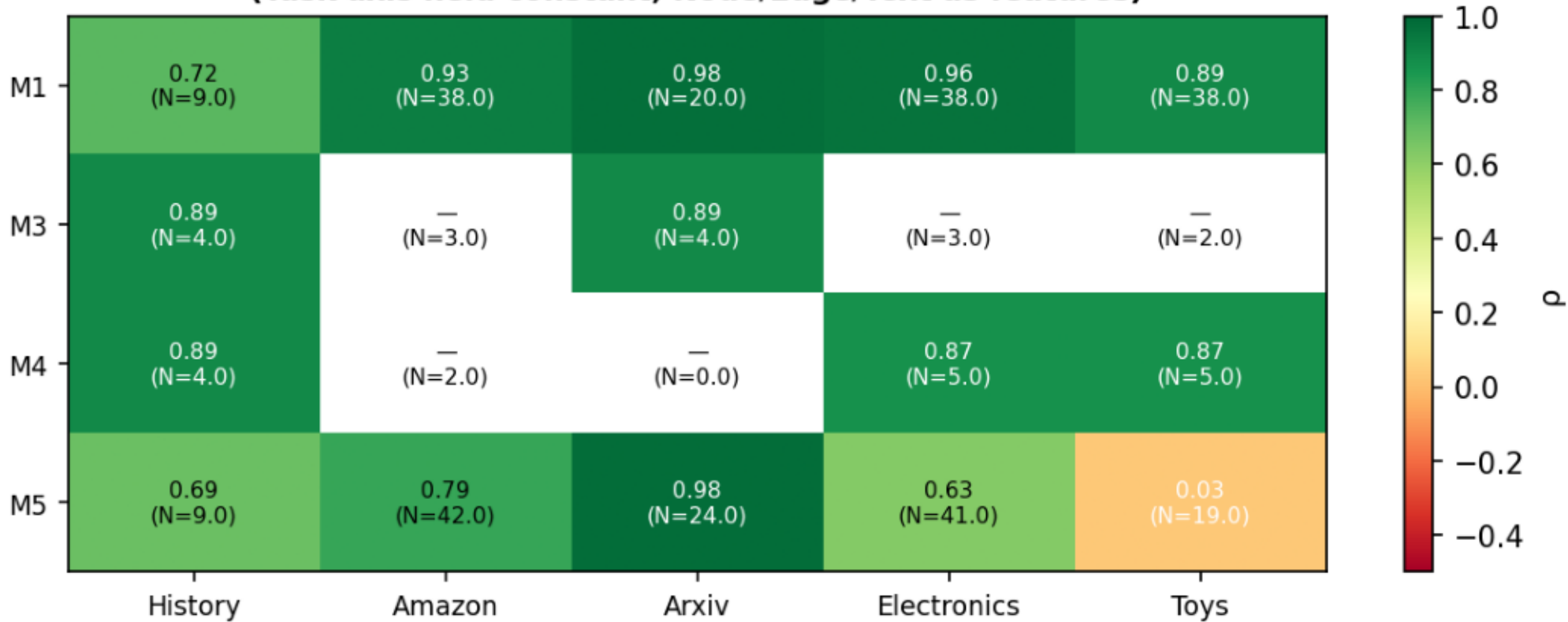
Check 2A — Task_Idx Ablation: ρ With vs Without Task Axis



Tree refit with Task_Idx removed (Node, Edge, Text only). Drop in ρ = variance explained by task-type identity. Note: Toys ρ increases without Task_Idx — task axis is actually hurting there.

Check 2B — Within-Task-Type Strategy-1 ρ (construction quality signal test)

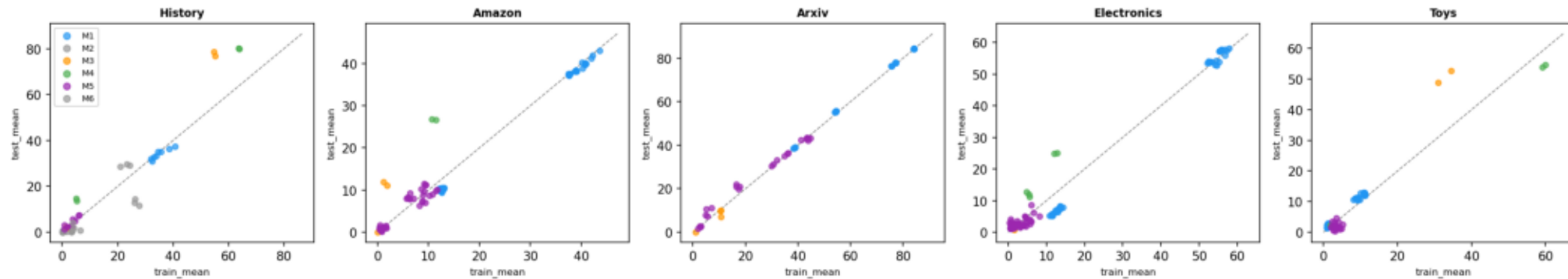
Check 2B — Within-task-type Strategy-1 ρ
(Task axis held constant; Node/Edge/Text as features)



Tree fit on Node/Edge/Text only, separately per task type (Task held constant).
M1 ρ reaches 0.961 (electronics) and 0.976 (arxiv). Toys M5 collapses to 0.029.

Check 3 — train_mean vs test_mean Scatter (colored by task type)

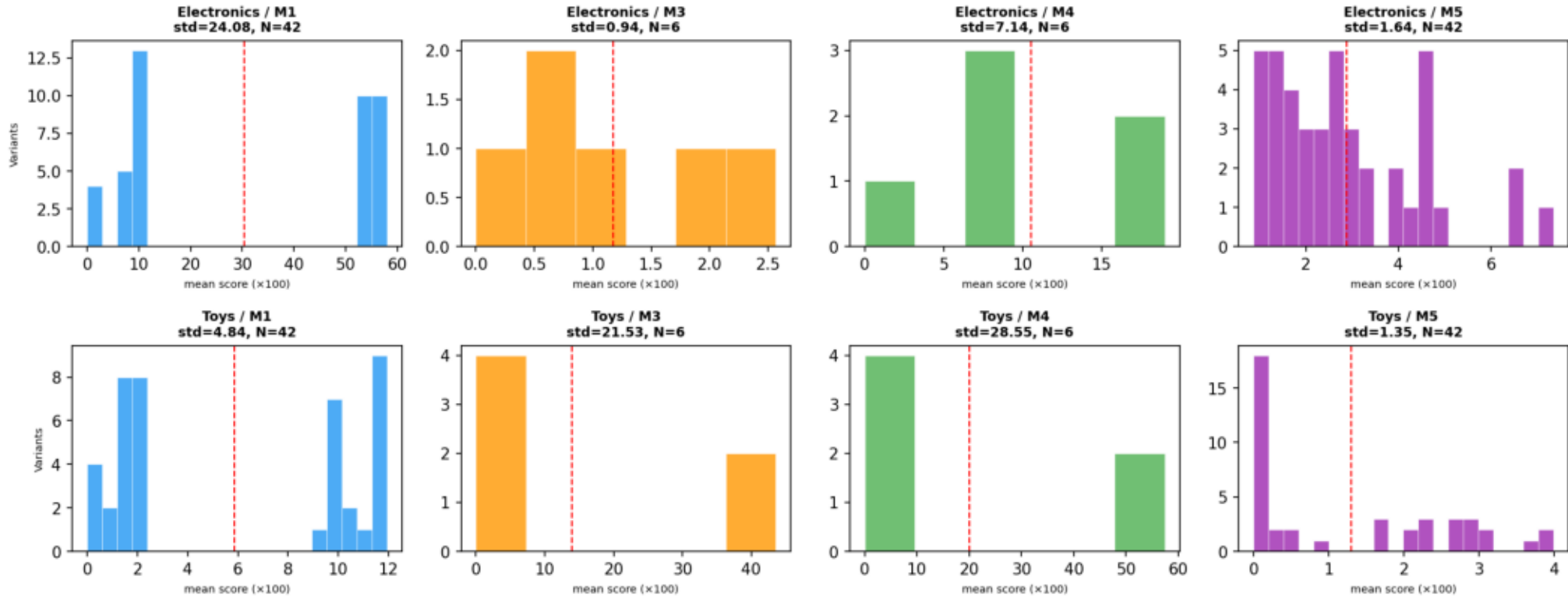
Check 3 — train_mean vs test_mean scatter (colored by task type)
Diagonal = perfect agreement; systematic offset by task type = confound



Each dot = one variant. Tight diagonal = real signal. Systematic vertical clustering by task type would indicate confound. Filter-excluded variants (zero-pool) all cluster at near-zero test_mean.

Check 5 — Per-Variant Score Distribution by Task Type (Electronics & Toys)

Check 5 — Per-variant mean score distribution by task type
(Electronics & Toys; red = mean; narrow spread = compressed signal)



Electronics M1: std = 24.1 across mean = 30.4 — good spread, construction choices matter.

Toys M1: std = 4.8, max = 11.9 — compressed. Toys M5: std = 1.35 — nearly flat signal.